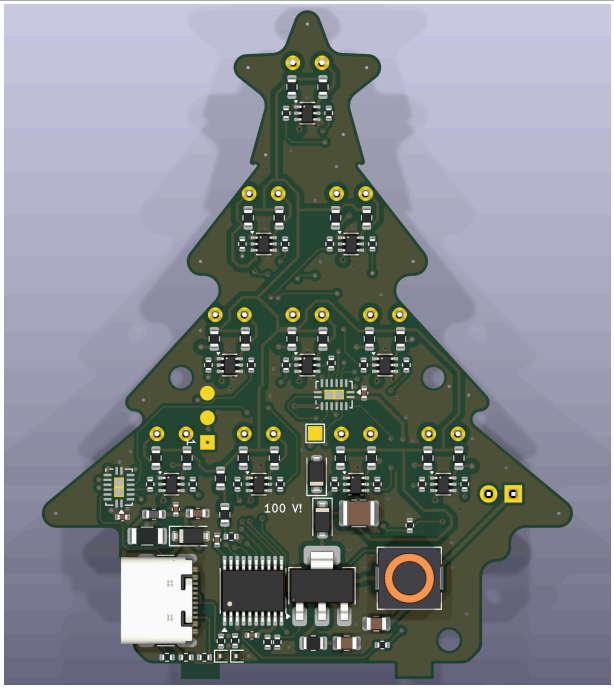
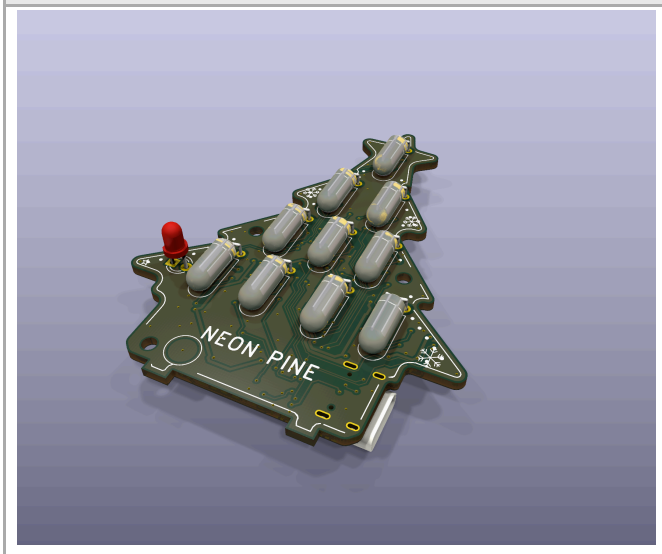
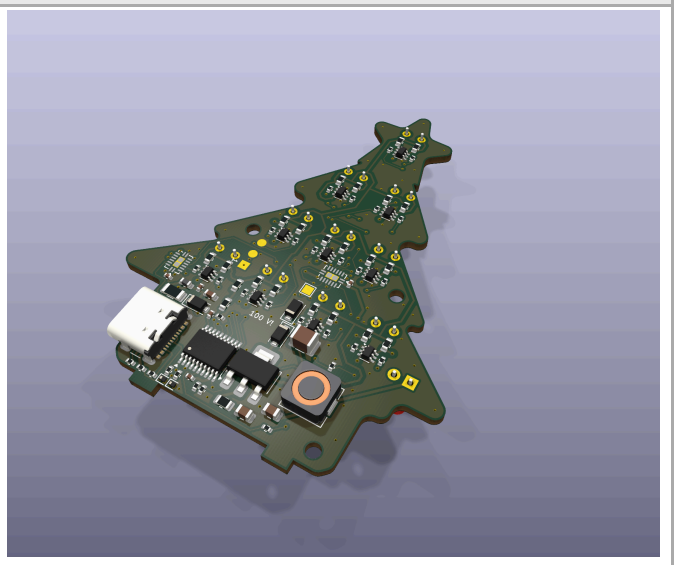
NeonPine – neon-lamp Christmas tree night light

54.8 × 69.1 mm tree-shaped PCB with 10 Brightmaster NEON-3 neon lamps, powered from USB-C 5 V.

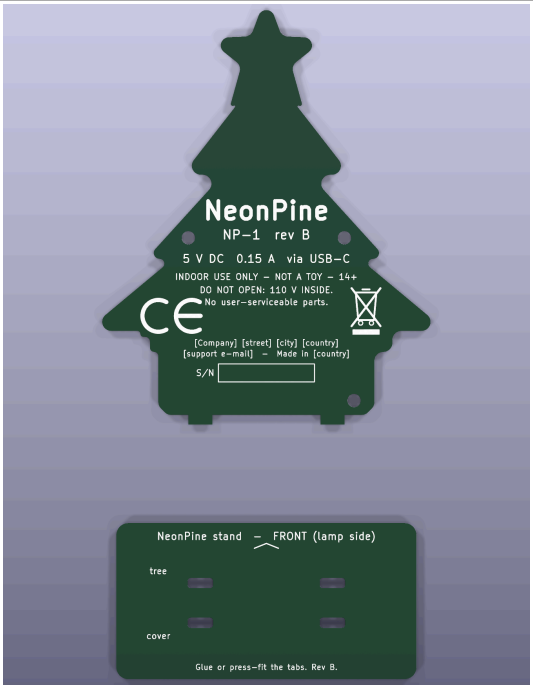
> **Branches `AC` + `USB` (2026-10-13): routed, 0 DRC errors.** Changes since rev B:
>
> - **Bipolar lamp drive:** each lamp sits in an H-bridge (§2.3), so either electrode can glow, both (alternating) or neither. Each lamp's switches are one **MMDT5551 dual 160 V NPN (SOT-363)** with 0603 HV pull-ups and 0402 base resistors; the low-voltage passives are 0402, the shift registers DHVQFN-16, D1/D2 SOD-123F, the board 2 layers.
> - **Low-speed USB** with rv003usb (§2.6): J1 is a GCT **USB4105-GF-A** (USB 2.0, 16P). The MCU stays at 5 V; 68 Ω series resistors and 3.6 V zeners clamp its D+/D− drive. This moved several MCU pins (§2.5) and needs **NRST configured as GPIO** (PD7 = SR_DATA).
>
> DRC: 0 errors, 0 unconnected, 0 schematic-parity issues, 5 silkscreen warnings (the same as before the USB work). Hand edits (J2, U3, Q15/R55, the whole USB area) live only in the board file – `tools/pcb.py` / `route.py` would not reproduce them. `docs/compliance/` still describes rev B. `production/` (Fabrication Toolkit, 2026-10-13 09:29) predates the bottom-silkscreen cleanup and the solid J1 shell pour – re-export it before ordering; `fab/` is current.

A CH32V003 MCU runs a 75–105 V boost converter and drives each lamp separately. Rev B adds a screwed-on rear cover (with the product markings) and a slot-in stand base. Unit manufacturing cost (rev B estimate) ≈ €19 / €9 / €6 at 10 / 100 / 500 pcs (manufacturing §7).

Front	Back (cover removed)
	

Front	Back (cover removed)
	

Rear cover (outer face) and stand base – `NeonPine-cover.kicad_pcb`



Status (2026-10-13): layout complete (H-bridge lamp drive + USB). Fully routed: 0 DRC errors, 0 unconnected items, 0 schematic parity issues. All rev A design actions that blocked market release are done in hardware (§7). **Still open before market release:** firmware, bench and safety verification, the EMC test campaign, supplier declarations, and the company details in the markings (regulatory roadmap).

Document	Contents
docs/NeonPine-schematic.pdf	Schematic (generated)
docs/README.md	Index of all product documentation
docs/compliance/	Regulatory roadmap, risk assessment, EMC test plan, materials, draft DoC / SDoC, labelling, technical file
docs/user-manual.md	User instructions and safety information (draft)
docs/manufacturing.md	PCB spec, assembly incl. cover & stand, programming, end-of-line test, traceability
docs/mcu-netlist.md	Reduced netlist for firmware: MCU pin map, 74HC595 bit order, lamp drivers, peripheral setup (generated)

Document	Contents
fab/	Main board: Gerbers + drill, pick & place, BOM, STEP. Cover + stand: Gerbers + drill

1. Concept

```
            ★   L1 hangs from the centre of the star
        L2   L3           tier 1
      L4   L5   L6        tier 2          ○ cover screw holes in the tier-2 wings
    L7   L8   L9   L10    tier 3          light sensor = "ornament" in the left wing
    ○    ▙   pot   ▟      "NEON PINE", trunk touch pad TP2, USB-C on the right edge
         ▙▄▄▄▄▄▄▄▟       two stand tabs → slot into the base plate
```

- **Lamps:** Brightmaster **NEON-3** (TME). Ø4 × 10 mm, orange, both wire leads out of one end, rated 65 V AC / **90 V DC** maximum striking voltage, 0.5 mA, 20 000 h. Brightmaster does not publish the maintaining voltage, so we assume ≈ 60 V as for an NE-2.
  - Footprint `NeonPine:Lamp_Neon_Brightmaster_NEON-3_D4x10_P3.0`: 3.0 mm lead pitch, and the body lies flat on the front, hanging down like an ornament.
  - Rev B: the star lamp LA1 hangs from the centre of the star like the others, so its glass stays on the board.
  - 3D model from `tools/lamp3d.py`.
- **Sides:** only the lamps, light sensor and trunk touch pad are on the front. Everything else is SMD on the back, under the rear cover.
- **Front art (silkscreen):** a rim line, bead "fairy lights" along the branches, snowflakes, stars and the NEON PINE logotype. It is clipped automatically around pads, holes and parts (`art()` in `tools/pcb.py`).
- **Rear cover:** a plain FR-4 plate in the tree shape, on 3 × M2 × 5 mm standoffs. It covers every HV part and carries the product markings in silkscreen: model, rating, CE, WEEE bin, warnings, manufacturer, S/N field.
- **Stand:** two 4 mm tabs on the bottom edge of the tree and of the cover plug into a 50 × 26 mm base plate. The base comes from the same cover/stand Gerber set.

2. Schematic (PDF)

2.1 Power in

```
USB-C J1 (GCT USB4105-GF-A) VBUS ─┬─ F1 PTC 0.5 A (MF-NSMF050-2) ── VBUS_F ── FB1 600 Ω@100 MHz ─┬─── +5V
CC1 ─R3 5k1─ GND, CC2 ─R4 5k1─ GND D3 SMF5.0A TVS (at the connector)              C1 10µ, C2 100n ┘
D+ / D─ → USB data network, §2.6
```

- +5V feeds U1 (C6 100n at VDD) and the boost (C3 22µ/10V). The ferrite and the caps form a π filter that keeps boost noise off the USB cable.
- The MCU runs straight from 5 V (the CH32V003 is rated 2.7–5.5 V), so the gate drive is a full 5 V.
- The expected maximum input is about 0.1 A.

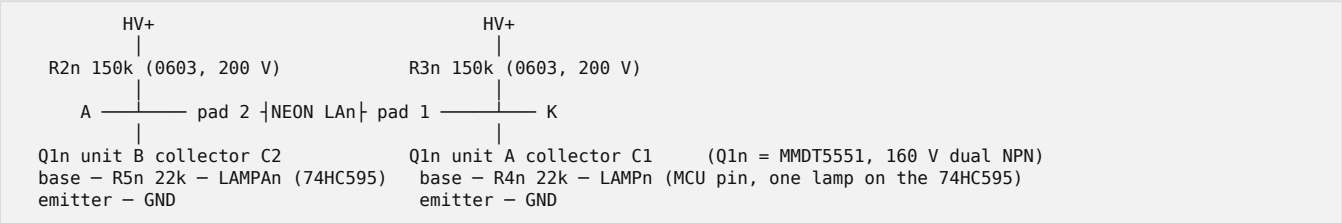
2.2 HV boost (the MCU is the controller)

```
+5V ── L11 330µH ──SW─┬─── D1 200 V fast ──┬──────────┬──────────┬── HV+ (75–105 V)
                      │   (R12 100Ω + C8   C4 1µ/250V D2 SMF100A  R5 1M ─ R6 1M ─┬─ VFB → PD3 (ADC A4)
                  Q1 DMN15H310SE  100p snubber, (1210)   (clamp)                 R7 39k ‖ C5 10n
PD6 ─R1 47Ω─ G    (150 V, 330 mΩ @ 5 V)  DNP)  │          │                       │
          R2 100k to GND                      GND        GND                     GND
```

- **Gate drive:** PD6 = TIM2_CH3 with the TIM2 full remap (AFIO TIM2_RM = 11); PD2, the old TIM1_CH1 pin, is USB D+ now.
- **Q1 DMN15H310SE:** 150 V, R_DS(on) 330 mΩ at V_GS = 5 V. **R2 is required**: it keeps the boost off during reset and programming. D2 (SMF100A, SOD-123F, 200 W) clamps HV+ at V_BR 111–123 V, below Q1's 150 V.
- **R12/C8:** an RC snubber across Q1, not fitted by default (DNP). Fit it if pre-compliance shows SW-node ringing at 30–100 MHz.

- **L11:** 330 µH shielded 6 × 6 × 2.8 mm (SWPA6028S331MT, **verify I_sat ≥ 0.4 A** on the datasheet). Footprint `NeonPine:L_6.3x6.3_H3`, 3D model from `tools/inductor3d.py`.
- **Design point:** 30 kHz in discontinuous conduction mode, P_out max ≈ 0.35 W, I_pk ≈ 0.27 A, t_on ≈ 18 µs at 105 V.
- **Feedback:** 2 M / 39 k gives 105 V → 2.0 V at the ADC. Calibrate against Vrefint. The divider also bleeds C4 after unplugging (τ ≈ 2 s, below 60 V in about 1.1 s).
- **Protection:**
  - Firmware: 60 % duty clamp, an overvoltage trip at 110 V, and the IWDG watchdog.
  - Hardware: R2 (gate pull-down), D2 (clamp), F1 (PTC), D3 (TVS).
  - Firmware limits are **not** counted as safety measures.
- **Stored energy:** at most ½ · 1 µF · 105 V² ≈ 5.5 mJ in C4.

2.3 Lamp channel (× 10) – H-bridge per lamp



Q1n B (A side)	Q1n A (K side)	Result
off	off	dark, no current
off	**on**	K is the cathode: the **pad-1 electrode** glows
on	off	A is the cathode: the **pad-2 electrode** glows
alternate ≥ 100 Hz		both electrodes glow (AC look); also evens out cathode wear
on	on	dark, but both pull-ups draw current – firmware never does this

- **Parts:** Q1n = MMDT5551 (Diodes Inc., V_CEO 160 V, SOT-363, pin-out E1 B1 C2 E2 B2 C1); 160 V covers the 123 V D2 clamp. The pull-ups dissipate up to 105² / 150 k = 74 mW (the lead that is pulled low), so they must be **0603 high-voltage thick film rated ≥ 200 V / 0.1 W** (e.g. RCV0603); a standard 0603 is only rated 75 V. 0402 is not possible here (≤ 100 V even in HV series). No 160 V+ dual NPN exists in a DFN package (Nexperia's DFN2020 doubles stop at 120 V, PBSS4112PAN), and there is no quad.
- Lamp current is I = (V_HV − 60 V) / 150 k through the pull-up of the lead that is *not* pulled low: 0.31 mA at 105 V, 0.11 mA at 75 V. **The HV rail works as a smooth global dimmer (about 3:1)**, and per-lamp BAM adds detail on top of it.
- **Power budget – the cost of this topology.** The pull-up on the lead that *is* pulled low also conducts, V_HV / 150 k = 0.5–0.7 mA, more than the lamp itself. One lit lamp therefore takes 1.0 mA × 105 V ≈ 105 mW from the rail (45 mW at 75 V); ten lit lamps take ≈ 1.05 W at 105 V. The rev B boost (§2.2) delivers ≈ 0.35 W (≈ 0.45 W in theory at the 60 % duty clamp), so **the firmware must hold Σ duty × P_lamp(V_HV) under about 0.35 W**, e.g. by lowering V_HV when many lamps are lit. For full brightness on all lamps the boost needs about 3× the power: a lower inductance with I_sat ≥ 1 A (e.g. 100 µH) plus a firmware frequency/duty retune – open item, not done in this branch.
- **Striking:**
  - Firmware starts the rail at ≥ 95 V (a kick to 105 V), and only ramps down once the lamps have struck.
  - A lamp that drops out below about 65 V is re-struck with the same kick.
- **Touch current at the lamp leads** is at most 105 V / 150 k = **0.7 mA DC**, because both leads are fed only through their 150 k pull-ups.
- With DC drive only the cathode glows; here the firmware chooses which lead is the cathode, per lamp and per BAM slot.
- **Dark effect:**
  - Firmware keeps BAM at 1 kHz or faster with a minimum on-time, plus a 50 ms kick at first ignition.
  - Lamps with a radioactive additive (Kr-85) must **not** be used.

2.4 Inputs

Input	HW	Use
TP2 trunk touch pad (front, Ø4 mm, **covered by solder mask**)	R11 1 k to PA2 (TKEY)	tap: next animation · hold: brightness ramp · double-tap: sleep timer · hold 3 s: off/on
Light sensor Q2 TEPT4400 (Vishay, Cd-free NPN phototransistor, front left wing)	+5V → C, E → PA1 (ADC A1), R9 22k to GND	auto on at dusk / off in daylight, room-adaptive brightness, wave-to-switch. 20 lx → 0.3–1.5 V, 100 lx → about 4.4 V
TP3 (back pad)	SR_CLK (PC7)	test point for the shift-register clock

Rev B removed the star touch pad TP1 and its resistor R10, and replaced the CdS LDR (GL5528, contains cadmium) with the TEPT4400. **Wave-to-switch:** the light sensor is read in the BAM slot where all lamps are dark and again in a lit slot; the difference is the tree's own light reflected off a hand.

2.5 MCU pin map – CH32V003F4P6 (TSSOP-20)

Full pin / net / bit tables for the firmware: docs/mcu-netlist.md (generated by `tools/mcu_netlist.py`).

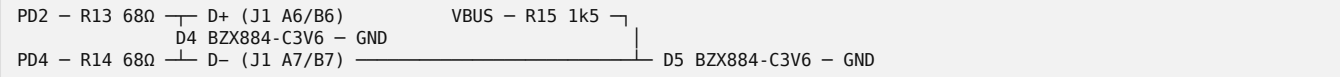
Pin (port)	Function
10–16 (PC0–PC6)	Cathode-side switches of lamps **L7, L4, L2, L5, L3, L9, L6**
8 (PD0)	Cathode-side switch of **L8**
19 (PD2) / 1 (PD4)	**USB D+ / D−** (rv003usb, through R13 / R14 68 Ω; D− = EXTI4)
4 (PD7) / 17 (PC7) / 2 (PD5)	**SR_DATA / SR_CLK / SR_LATCH** to the 74HC595 chain U2 → U3 (TP3 sits on SR_CLK). **PD7 is the NRST pin: program NRST as GPIO** (`minichlink -D`)
3 (PD6)	TIM2_CH3 (TIM2 full remap) → boost gate (via R1)
20 (PD3)	A4 – HV feedback
5 (PA1)	A1 – light sensor
6 (PA2)	A0 – TP2 trunk touch
18 (PD1)	SWIO, on pogo pads J2 (5V / SWIO / GND), back side

If NRST is left enabled, PD7 cannot drive SR_DATA (the 595s then shift in ones and the lamps light wrongly), but nothing unsafe happens.

Shift registers (2 × 74HC595BQ, DHVQFN-16 2.5 × 3.5 mm; exposed pad = GND (footprint `NeonPine:DHVQFN-16-…_EPGND`, EP numbered 8)): U2 QA–QH = L8A, L7A, L4A, L2A, L1A, **L1 (cathode side)**, L5A, L3A; U3 QD = L10A, QE = **L10 (cathode side)**, QF = L9A, QH = L6A (QA–QC, QG unused). /OE is tied low and /SRCLR high: the outputs are random until the first latch, which is harmless because the boost is held off by R2 until the firmware runs. One 16-bit shift + latch per BAM slot (8 slots per ≈ 1 ms frame) is a few µs of CPU time.

The lamp-to-pin order (MCU pins and shift-register outputs) comes from `tools/pinswap.py` (stored in `tools/lamp_pins.json`); the USB rework moved L10's cathode switch to U3 QE and L10A to U3 QD by hand (rerunning `pinswap.py` would undo that). **The firmware maps lamp index → port bit through a table.**

2.6 USB (low speed, rv003usb)



- **Why the MCU can stay at 5 V** (full gate drive for Q1, no LDO, no level shifter): at VDD = 5 V the CH32V003's V_IH is 0.22·(VDD − 2.7) + 1.55 = 2.06 V, so it reads the 3.3 V bus directly. When it transmits, its 5 V push-pull output is clamped by the 68 Ω + 3.6 V zener pair (≈ 20 mA peak per line for the length of a packet).
- **D− pull-up** R15 1k5 to VBUS → low-speed device, **always on the bus**; D5 clamps the idle level. While plugged in it draws ≈ 1 mA through D5. To re-enumerate (bootloader → application), drive

D− low for a few ms.
- **rv003usb** `usb_config.h`: `USB_PORT D`, `USB_PIN_DP 2`, `USB_PIN_DM 4`, no `USB_PIN_DPU`. Needs HSI × 2 = 48 MHz.
- **Layout:** J1 = GCT USB4105-GF-A (LCSC C3020560). Its shell front sits 1.64 mm past the board edge (x 119.6, 1.2 mm further out than GCT's PCB-edge line), and the right pot-rim lip above it is cut back to x 119.4 (`outline()`; the rear cover keeps the earlier, smaller notch, which only cuts the lip below y 97 and has no effect on the cable): the plug body of a cable (≤ 12.35 mm wide, reaching up to 1.6 mm forward of the back face) must not hit the board in front of the mouth – now clear for plug bodies up to 12.5 mm wide. R13/R14 under U1's pin-1 corner, D4/D5, R15 and the CC resistors R3/R4 in a row below J1. Its shell pads are poured solid. The library land pattern puts J1's A1/B12/A12/B1 pads 0.19 mm from its own NPTH pegs – a J1-only rule in `NeonPine.kicad_dru` allows that.

3. Firmware architecture (ch32v003fun or WCH SDK) – *not written yet*

- **Clock:** 48 MHz from HSI × 2 (also required by rv003usb). **TIM2_CH3** (PD6, TIM2 full remap) runs the 30 kHz boost PWM with **±5 % frequency dithering** (EMC measure E2).
- **Option bytes:** NRST disabled (PD7 = SR_DATA). If the rv003usb bootloader is used, keep RST_MODE = GPIO when configuring it.
- **USB:** rv003usb (HID, no driver needed); optional rv003usb bootloader for firmware updates over USB. Its EXTI4 (D−) interrupt must have the highest priority and must never be preempted: keep other ISRs short (≈ 40 cycles) or preemptible.
- **Boost loop:** a 1 kHz PI loop driven by the ADC, ramping at most 2 V/ms; the setpoint is the global brightness, 75–105 V. When everything is dark: PWM off and MCU standby.
- **Lamp PWM:** 8-bit bit-angle modulation from TIM1 with an ≈ 1 kHz frame and gamma correction.
- **Animation engine:** each mode is a `frame(t) → uint8_t lvl[10]` function, and modes crossfade over 1.5 s.
- **Single-pad UI on TP2:** tap / hold / double-tap / long hold.
- **Compliance hooks:** an EMC worst-case mode (all lamps on, 105 V), a production self-test mode, and the firmware version readable over SWD and USB.

#	Mode	Idea
1	Candle	independent 1/f flicker per lamp
2	Breathe	all lamps 100 %, only the HV rail moves
3	Twinkle	random lamps swell and decay
4	Garland	spiral chase up the tiers, ending in a star flash
5	Snowfall	soft pulses drifting top → bottom
6	Heartbeat	steady tree, the star double-pulses
7	Drift	auto-cycles 1–6 every 3 min
8	Sleep	dims over 20 min, then off

4. BOM (approx. LCSC prices; total last checked for rev B)

Full list: fab/NeonPine-bom.csv. R12 and C8 are DNP.

Ref	Part	Qty	≈ $
U1	CH32V003F4P6	1	0.13
LA1–LA10	Brightmaster NEON-3 (TME)	10	*TME price*
Q11–Q20	MMDT5551 (dual NPN, SOT-363)	10	≈ 0.20 *(check)*
U2, U3	74HC595BQ (DHVQFN-16)	2	≈ 0.10 *(check)*
Q1	DMN15H310SE	1	0.15
Q2	Vishay TEPT4400 (light sensor)	1	0.10

Ref	Part	Qty	≈ $
L11	SWPA6028S331MT 330 µH	1	0.06
D1 / D2 / D3	200 V 1 A fast diode, SOD-123F *(pick P/N)* / SMF100A / SMF5.0A	1+1+1	0.07
C4	1 µF 250 V X7R 1210	1	0.06
R21–R40 / R41–R60	150k 0603 HV (≥ 200 V) / 22k 0402	20+20	≈ 0.15 *(check HV part)*
other R	0402 (47, 100k, 5k1 ×2, 39k, 22k, 1k, 68 ×2, 1k5); R5/R6 1M stay 0603 (≈ 52 V across each, 0402 is rated 50 V)	12	0.02
D4, D5	BZX884-C3V6 3.6 V zener, SOD-882 (LCSC C7529010)	2	0.02
C1–C6, C9, C10	0402: 100n ×4, 10n · 0603: C1 10µ · 0805: C3 22µ/10V · 1210: C4 1µ/250V (bulk caps kept larger for DC-bias derating)	8	0.05
FB1	600 Ω @ 100 MHz 0603	1	0.01
J1 / F1	USB4105-GF-A (USB 2.0, LCSC C3020560) / MF-NSMF050-2	1+1	*check*
mech	rear cover + stand PCB, 3 × M2 × 5 standoff, 6 × M2 × 4 screw	–	*PCB price* + 0.10
	Total excl. lamps and PCBs (rev B, live LCSC prices 2026-10-06; **re-quote** for the H-bridge + USB BOM)		**$3.17 / $2.47 / $2.15** at 10 / 100 / 500 units

5. PCB & layout

- **Board: 2 layers** (F.Cu / B.Cu = all SMD parts; was 4 layers, reduced for cost), FR-4 1.6 mm, 54.8 × 69.1 mm incl. stand tabs (`SC` in `tools/pcb.py` can scale the whole tree if needed). The HV supply (HVBUS) runs on B.Cu only, never as a track on F.Cu. Verified 2026-10-13: **0 HVBUS vias and 0 HVBUS tracks on F.Cu**, so the front carries no unlimited-HV copper. GND pour on F.Cu and B.Cu (SMD pads solid, THT pads with thermal spokes, USB shell solid); all vias tented.
- **Netclasses** (`tools/pcb.py` writes them to `NeonPine.kicad_pro`):

Class	Nets	Track	Clearance	Via	Layers
HVBUS	HV+, SW, snubber node Net-(C8-Pad1) (unlimited 105 V source)	0.3 mm	**0.5 mm**	–	**B.Cu only** (enforced in `route.py`, verified: 0 items on the front)
HV	lamp anode / cathode nets, FB_MID (current-limited through ≥ 150 k)	0.3 mm	0.5 mm	0.7/0.35	both
PWR	VBUS, VBUS_F, +5V, GND	0.4 mm	0.15 mm	0.7/0.35	both
Default	everything else (USB data included)	0.2 mm (0.15 min)	0.15 mm	0.6/0.3	both

  `NeonPine.kicad_dru` relaxes the clearance to 0.2 mm **inside the courtyards of Q11–Q20 only**: the SOT-363 itself puts its collector pins 0.3 mm from base/emitter. Freerouting does not read custom rules, so expect to hand-route the collector/base fan-outs (or let Freerouting route the rest and finish these by hand). A second rule lets J1's own pads sit 0.19 mm from its NPTH alignment pegs (§2.6).

- **Placement:**
  - Fixed by hand: the lamps and their R/R/Q stacks, J1, U1, Q1, L11, the input protection D3/F1 (right above USB-C), the 3 cover holes and the touch pad.
  - The other ≈ 25 parts are placed by the annealing optimiser in `pcb.py`, with weights on the boost loops, decoupling, the ADC filter and the USB input.
  - Results: D1 3.2 mm from Q1's drain tab, D1–C4 3.2 mm, C6 at U1 VDD 2.2 mm, the TVS 5.7 mm from VBUS.
- **Routing:** Freerouting **2.4.1** (`tools/route.py`; version 2.1.0 writes incomplete sessions). Freerouting leaves the last 2–3 tight connections open; `tools/fixup.py` closes them with short legal links

checked by KiCad's own collision test. The USB area (J1, U1 surroundings) was re-placed by `tools/usb_rework.py` and routed by hand; `tools/route_local.py` runs Freerouting on selected nets only, with every other track locked.

- **Design flow:**
  1. `tools/gen.py` writes the schematic, then `tools/sch_boost.py` redraws the boost section with real wires.
  2. `kicad-cli sch export netlist -o np.net NeonPine.kicad_sch` produces the netlist.
  3. `tools/pcb.py np.net` builds the outline, art, placement and pours.
  4. `tools/pinswap.py` solves the lamp-to-pin map; if it changed, repeat steps 1–3.
  5. `tools/route.py <freerouting-2.4.1.jar>` routes the board.
  6. `tools/fixup.py` closes the remaining links; then run DRC.
  7. `tools/cover.py` writes the cover and stand board.
  8. 3D models come from `tools/lamp3d.py` and `tools/inductor3d.py` (FreeCAD).
  9. `tools/docs.py` regenerates the schematic PDF, the renders, the firmware netlist (`tools/mcu_netlist.py`), the `fab/` outputs and the PDFs of the Markdown documents (compliance excluded).

6. Ratings (to be confirmed by measurement)

Parameter	Value
Input	5 V ⎓ via USB-C, ≤ 0.15 A (expected ≈ 0.1 A max, standby ≈ 2 mA incl. ≈ 1 mA in the USB D− pull-up)
Internal voltage	75–105 V DC, ≤ 123 V under fault (D2 clamp), only on the back under the cover
Environment	indoor, dry, 0–35 °C, IP20
Light output	≪ 60 lm (outside the scope of Ecodesign Reg. (EU) 2019/2020)

7. Market-release items

#	Item	Status
1	Rear insulating cover over all HV parts, tool-removable (3 × M2 screws)	**done in rev B** – verify with the IEC test probe and a drop test
2	HV supply (HV+/SW) on the back layer only	**done** – 0 HVBUS copper on the front, no HVBUS vias
3	Lamp supplier declarations: RoHS, REACH, **no Kr-85**	open (purchasing)
3b	Replace the CdS LDR with a Cd-free sensor	**done** – Vishay TEPT4400, R9 → 22k
4	Firmware with an EMC worst-case mode and a production test mode; EMC tests	open
5	Product markings	**done on the cover silkscreen** – fill in company, address and country; the S/N field takes a printed label
6	Lamp retention (silicone dab)	process step in `docs/manufacturing.md`
7	ESD: masked touch pad, TVS on VBUS	**done** (TP2 masked, D3 SMF5.0A); star touch pad removed
8	Part numbers for L11, F1, J1	**chosen** (SWPA6028S331MT, MF-NSMF050-2, USB4105-GF-A) – confirm L11 I_sat
8b	USB: enumeration on Windows / macOS / Linux hosts, D+/D− levels with the zener clamps, EMC with a data cable	open (bench)
9	Stand	**done** – tabs + base plate (cover/stand board)
10	Bench verification: D2 clamp vs Q1, C4 discharge, input current, temperature rise, touch current	open (risk assessment §5)
12	Price check findings: L11 → SWPA6045S331MT (the 6028 is rated only 300 mA and out of stock), C4 → 1812 footprint (no 1 µF/250 V in 1210), Q1 low stock → CJT04N15 alternative	open – see manufacturing §7.3

#	Item	Status
11	EMC extras: ferrite (done), snubber footprint (done, DNP), boost dithering (firmware), edge stitching vias (not done; add if pre-scan needs it)	partly